FCDNet: A Multiscale Attention Network for Forest Change Detection Using Dual-Temporal Very-High-Resolution Remote Sensing Images

Jun Wang, Zongqi Yao, Long Chen, Ruijing Yang, and Xiaoli Zhang, *Member, IEEE*

***Abstract*—Forest change detection (CD) plays a vital role in remote sensing research, serving as a cornerstone for ecological protection and sustainable environmental management. While existing research on CD primarily focuses on urban areas and croplands, forest CD remains underdeveloped. As the demand for real-time monitoring of forest changes grows in response to challenges like climate change and environmental degradation, advancements in remote sensing technologies make it possible to bridge this research gap. The complexity of forest CD stems from high seasonal and interannual variability, with fluctuations often showing strong similarities across years. In this article, we introduce *FCDNet: A Multiscale Attention Network for Forest CD Using Dual-Temporal Very-High-Resolution Remote Sensing Images*. The proposed network is designed to enhance the precision and robustness of CD through the integration of multiscale feature aggregation and an adaptive channel attention strategy. This design enables FCDNet to sensitively identify fine-grained forest alterations while maintaining strong adaptability to spatial and temporal variability in the input imagery. Comprehensive experiments confirm that FCDNet delivers superior performance compared with existing state-of-the-art approaches, showing notable advantages under challenging scenarios characterized by seasonal transitions and interannual fluctuations. FCDNet achieves impressive $F1$-scores of 78.51%, 90.48%, and 92.57% on the Forest CD dataset (FCDD), LEVIR-CD, and WHU-CD datasets, respectively, delivering state-of-the-art results. This approach holds significant potential for advancing forest monitoring, providing a reliable tool for researchers and policymakers to better understand and protect forest ecosystems.**

***Index Terms*—Channel exchange, deep learning, forest change detection (CD), multiscale attention, remote sensing.**

I. Introduction

FORESTS, as one of Earth's most vital ecosystems [1], play a key role in regulating atmospheric CO_2 levels, maintaining climate balance, and supporting biodiversity. Forest cover change is a crucial indicator of global environmental dynamics, with far-reaching ecological and climatic implications. However, large-scale deforestation and forest degradation—driven by human activities, climate change, pests, and diseases [6]—pose serious threats to ecosystem stability. The FAO reports an annual global loss of approximately 70 million hectares of forest [2], a trend that could cause irreversible ecological damage if left unchecked [3].

As the largest terrestrial carbon sink [4], forests are central to greenhouse gas regulation. Monitoring forest changes is thus essential for assessing carbon cycle dynamics and shaping effective carbon reduction strategies [5]. Moreover, these changes often manifest gradually and heterogeneously, requiring not only timely detection but also continuous multitemporal analysis [7] to capture complex spatiotemporal forest dynamics.

Within the field of change detection (CD) [8], advancements in neural network methodologies, combined with the growing accessibility of very-high-resolution imagery, have significantly improved the precision and speed of identifying temporal changes. Traditional methods, often based on image discretization, struggle with subtle changes, complex backgrounds, and both low- and high-resolution data. These limitations are especially evident in forest CD, where models must handle multiscale features, class imbalance, and fine-grained changes.

Remote sensing CD involves identifying changes across time-series images. However, traditional postclassification comparison approaches suffer from error accumulation and often miss contextual background information, particularly in complex scenes. Deep learning methods have improved this by enabling networks to learn hierarchical multiscale features directly from raw data, improving both accuracy and robustness.

Among them, convolutional neural networks (CNNs) [9] remain widely adopted. CNNs extract spatial feature hierarchies through layered convolutions and are typically trained end-to-end [10], eliminating manual feature design. However, their performance can degrade in high-resolution imagery with noisy backgrounds or when detecting small, scattered changes, which remain challenging.

As illustrated in Fig. 1, the central goal of CD is to identify pixel-level transitions, where altered regions are represented in white and stable areas are shown in black. Generally,

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Jun Wang, Zongqi Yao, Ruijing Yang, and Xiaoli Zhang are with the State Key Laboratory of Efficient Production of Forest Resources, Beijing Key Laboratory of Precision Forestry, and the Key Laboratory of Forest Cultivation and Protection (Ministry of Education), Beijing Forestry University, Beijing 100083, China (e-mail: wangjun1704@bjfu.edu.cn; yaozq@bjfu.edu.cn; yangruijing2021211@bjfu.edu.cn; zhang-xl@263.net).

Long Chen is with the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing 100091, China (e-mail: chenlong@ifrit.ac.cn).

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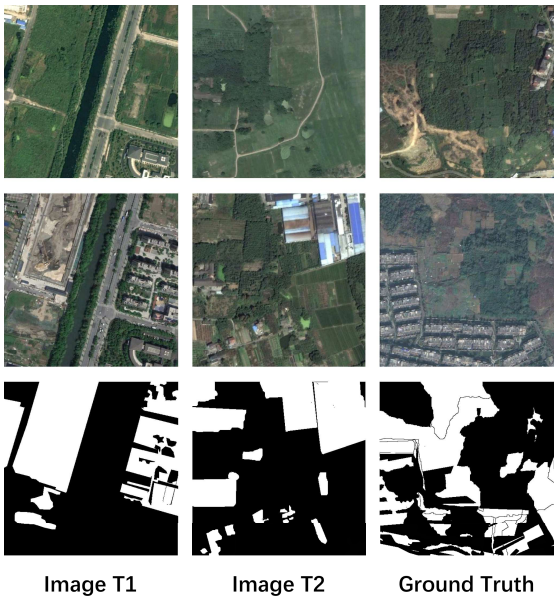


Fig. 1. Samples of the FCCD. White areas represent changes, and black areas represent unchanged areas.

VHR image CD is categorized into two types: foreground and background. The foreground represents meaningful changes, typically linked to regions of interest, such as buildings. In contrast, the background refers to changes that are not relevant to the area of interest, often arising from seasonal variations, lighting conditions, environmental factors, or other sources of interference.

To overcome limitations in CNN-based CD, advanced architectures such as fully convolutional networks (FCNs) [11] and U-Net [12] have been proposed. FCNs enable pixel-level prediction by combining convolution, pooling, and upsampling layers to capture both fine details and context. However, upsampling often leads to resolution loss, especially around edges, reducing detection accuracy in high-resolution images.

The U-Net architecture extends traditional convolutional models through an encoder–decoder design complemented by skip connections, enabling the retention of fine-grained spatial details and thereby improving the precision of CD. Yet, U-Net still faces challenges in fusing multiscale features, which limits its ability to detect changes of varying sizes effectively.

To prioritize critical regions in the imagery and mitigate variability from lighting conditions, seasonal changes, and intricate forest patterns, attention mechanisms [13] are employed. Variants like spatial attention [14] emphasize important spatial areas, while channel attention highlights key feature maps. Combining both helps models ignore irrelevant regions and improve subtle CD.

Handling multiscale information remains critical since changes range from small local alterations to large-scale shifts. Traditional multiscale extraction methods (e.g., pyramids and sliding windows) [15] can cause information loss, missing subtle changes. Multiscale feature fusion techniques [16], such as feature pyramid networks (FPNs) [17], combine features at different scales while preserving spatial and semantic details, improving robustness in high-resolution remote sensing.

Despite these advances, forest CD is underexplored. Existing methods [19], [20] struggle with subtle, seasonal changes and distinguishing forest from similar nonforest land covers, leading to misclassification and reduced accuracy. Therefore, new approaches are urgently needed to capture the complex heterogeneity and subtle dynamics of forests.

This article proposes *FCDNet*, a multiscale attention network designed for dual-temporal very-high-resolution forest CD. By integrating multiscale feature fusion and channel exchange mechanisms, FCDNet aims to enhance accuracy and robustness across diverse forest change patterns and temporal scales.

Therefore, the contributions of this article can be summarized as follows.

1) We propose FCDNet, a novel architecture for CD in remote sensing imagery that combines multiscale feature fusion, hierarchical attention, and a channel exchange mechanism. This configuration allows the network to extract detailed spatial information while also retaining comprehensive semantic context.
2) The FCDNet architecture integrates a channel enhancement (CE) module along with a channel exchange module to better capture fine-grained variations present in the input images.
3) Comprehensive evaluations across several benchmark datasets, such as Forest CD dataset (FCDD), LEVIR-CD, and WHU-CD, indicate that FCDNet surpasses current leading methods in key metrics, including Precision, Recall, IoU, Accuracy, and $F1$-score. Additional ablation experiments highlight the individual contributions of each module, confirming the overall stability and reliability of the proposed architecture.

This article is structured into five main sections. Section II details the data sources, network architecture, and training procedures. Section III outlines the experimental setup and summarizes the results. Section IV offers an in-depth analysis and discussion of the findings, and Section V concludes the study while highlighting potential avenues for future work.

II. Materials and Methodology

In this section, we provide descriptions of the datasets used and the architecture of our proposed model. Both are described in detail.

A. Materials

1) FCDD: This article uses the Semantic Change etection Dataset (SECOND) [32] as the base to generate a dedicated FCDD for the task of forest CD. The SECOND dataset is a publicly available semantic CD dataset containing pairs of aerial images from multiple cities. It includes 4662 image pairs, each 512×512 pixels, collected from high-resolution platforms, including optical and SAR sensors. For our forest CD task, we focus on image pairs with large forest areas, merging them into a "forest green land" category. The dataset includes pixel-level annotations, manually labeled by experts, ensuring accurate ground truth (GT). It covers six primary land

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TABLE I
SUMMARY OF THE DATASETS USED IN THIS STUDY

Dataset	Source	Main Target	Change Type	Annotation	Image Size
FCDD (Ours)	Based on SECOND	Forest change	Seasonal + Human-induced	Manual (pixel-level)	256×256
LEVIR-CD	LEVIR	Building change	Urban development	Manual (pixel-level)	1024×1024
WHU-CD	WHU	Building change	Urban reconstruction	Manual (pixel-level)	512×512

cover classes involving both natural and man-made changes. These classes include the following:

1) Nonvegetated ground (abbreviated as n.v.g. surface);
2) Trees;
3) Low-lying vegetation;
4) Water bodies;
5) Buildings;
6) Recreational areas (e.g., playgrounds).

The "nonvegetated ground" refers to impervious surfaces such as roads, pavements, and bare land. For our specific forest CD task, we are mainly concerned with green land and forest areas, which include tree canopy and low-lying vegetation *These two categories are merged into the "forest green land" category for the purpose of this study*.

The dataset is split into training and testing sets at a 4:1 proportion, allocating the majority of samples for model training while keeping a fifth for performance evaluation. The dataset is designed to reflect the real-world distribution of land cover categories during change periods. The selection of image pairs is randomized to avoid any bias, and the changes in the dataset include both seasonal variations and those resulting from urban development activities.

To ensure compatibility with our model and considering computational resource limitations, each image is resized to a uniform size of 256×256 pixels.

To *further validate the outstanding performance* of FCDNet in remote sensing image CD tasks, we conducted extensive experiments on two public CD datasets, LEVIR-CD [30] and WHU-CD [31].

2) LEVIR-CD: The LEVIR-CD dataset is a large-scale remote sensing building CD dataset containing high-resolution images from multiple urban areas. With change pixels accounting for only 4.65% of the total pixels, and including complex seasonal and lighting variations, this dataset poses significant challenges for CD. We adopted the default data split provided by the authors and compared the performance of FCDNet with several state-of-the-art methods.

3) WHU-CD: The WHU-CD dataset is another challenging remote sensing building CD dataset, where change pixels account for only 4.26% of the total pixels. The dataset covers several urban areas, with changes mainly occurring in urban reconstruction and expansion. Similar to the LEVIR-CD dataset, WHU-CD also suffers from a highly imbalanced distribution of change and unchanged pixels, which further complicates the CD task.

To give a clearer overview of the datasets used in this study, we summarize their key properties in Table I, including the source, primary target, type of change, annotation level, and image size.

B. Methods

1) Baseline Model: Our model framework builds on a partial VGG16 [21] network as the encoder, with its input layer replaced by an adaptive convolution layer to accommodate different types of upstream data. We choose VGG16 due to its stable performance, simpler structure, and compatibility with CD tasks where shallow-to-mid-level features are more crucial. To ensure efficient and accurate feature extraction for dual-time-series remote sensing images, we also refer to existing frameworks such as C2FNet [22] and HANet [23] and propose a design that integrates multiscale feature fusion and a feature-guided decoder.

Specifically, our model first uses an adaptive convolution layer to process the input images, ensuring the model can handle input data of varying shapes. Subsequently, features are extracted at multiple scales using a dedicated module, enabling the model to better represent both fine and coarse details and thereby improving CD accuracy. Additionally, a representation-guided decoder is employed to fuse multiscale feature information, ensuring that the final CD results are more accurate and reliable.

2) Proposed Multiscale Attention Network for Forest CD: As illustrated in Fig. 2, our proposed model, *FCDNet*, is composed of several core modules that collaboratively address the challenges of forest CD in very-high-resolution remote sensing images. Each module is designed with a specific purpose to enhance feature extraction, representation, and fusion, aiming to improve both sensitivity and robustness in detecting complex forest changes.

A central innovation of FCDNet is the *multiscale feature fusion module*. Forest changes manifest at various spatial scales—from small, localized disturbances such as tree removal or disease outbreaks, to large-scale regional transformations like deforestation or land use change. Traditional CD approaches often operate at a single spatial scale, limiting their ability to capture such diverse change patterns. Our multiscale fusion (MF) strategy processes image features at multiple resolutions simultaneously, integrating fine-grained details with broader contextual information. This allows FCDNet to more effectively detect subtle local changes while maintaining awareness of large-area forest dynamics, significantly boosting detection accuracy and reliability.

Another key component is the *channel exchange mechanism*, which dynamically facilitates information sharing between different feature channels. Forest ecosystems exhibit complex spatial structures and temporal dynamics influenced by multiple environmental factors. Conventional feature extraction methods tend to treat channels independently, neglecting potential correlations. The channel exchange

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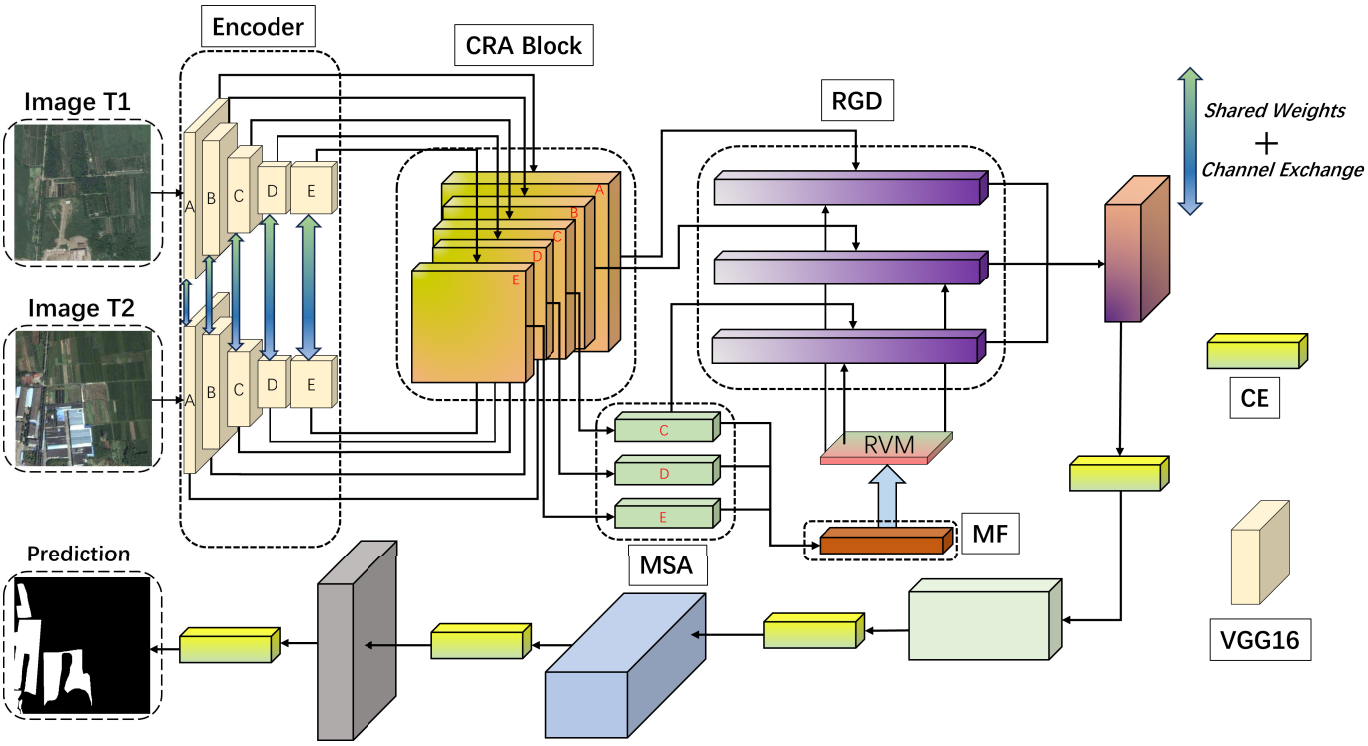


Fig. 2. Frame of the proposed FCDNet module.

mechanism in FCDNet addresses this by enabling selective communication and fusion across channels, enhancing the network's ability to represent heterogeneous forest features and temporal variations. This mechanism leads to a richer and more discriminative feature representation, directly improving the model's capacity to differentiate subtle changes from background variations.

Additionally, FCDNet employs attention mechanisms that further refine feature maps by focusing computational resources on the most informative spatial regions and channels. This attentional focus helps suppress irrelevant background noise caused by seasonal changes, lighting differences, or environmental factors, which are common challenges in forest monitoring.

Collectively, these modules establish a unified architecture capable of handling the complexities inherent in forested environments, achieving performance improvements over current state-of-the-art methods. Sections II-B3–II-B6 provide a detailed description of each module and its contribution to the overall network.

3) Encoder: The encoder serves as the first stage of the network, tasked with extracting meaningful features from the input images. To accommodate dual-temporal remote sensing inputs, the first convolutional layer of VGG16 is replaced by an adaptive convolution layer, which can process input tensors of varying shapes and resolutions flexibly.

Following this, the encoder includes four downsampling modules based on the VGG16_BN architecture. These modules sequentially reduce spatial dimensions while extracting hierarchical spatial features. Specifically, the adaptive convolution block is denoted as module A, while modules B to E correspond to VGG16 layers 2-12, 12-22, 22-32, and 32-42, respectively. This arrangement allows the encoder to produce multilevel feature representations that capture both low-level and high-level spatial information. The extracted features are then forwarded to the multiscale feature extraction modules for further processing.

4) Multiscale Feature Fusion: To improve the representation of change-related information across multiple spatial scales, the model integrates a multiscale feature extractor that captures both coarse and fine-grained contextual patterns. This design helps mitigate issues caused by scale variations and improves the detection of small and irregular changes.

The effectiveness of multiscale feature extractors has been widely acknowledged in recent remote sensing studies [43], [44], [45], [46], which demonstrate their ability to improve feature representation and model accuracy across various tasks. Our approach aligns with these findings by fusing features from multiple resolutions and incorporating adaptive attention to further refine the representation.

a) Channel Exchange Mechanism: The channel exchange mechanism is designed to facilitate information sharing between different channels within a feature map, aiming to enrich feature representations. As depicted in Fig. 3, the channel exchange mechanism partitions the input feature map into several groups according to a predefined hyperparameter. Within each group, features remain unchanged, while features are exchanged between different groups. This process enables each channel to incorporate information from other groups, allowing the network to capture multiscale and contextual relationships more effectively.

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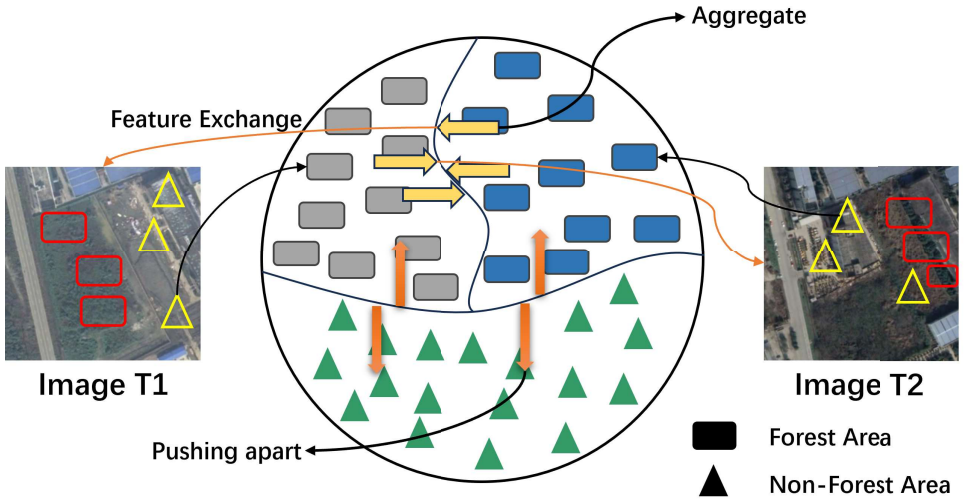


Fig. 3. Overview of the proposed channel exchange module.

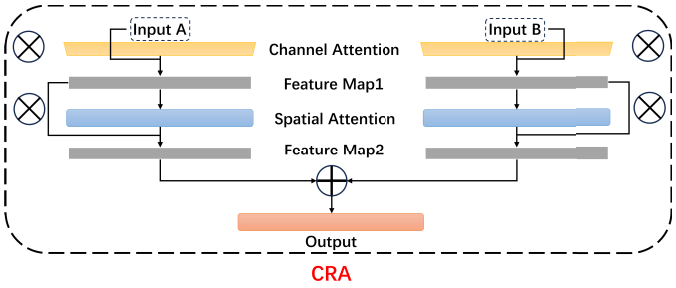


Fig. 4. Illustration of the proposed CRA module.

For dual-temporal remote sensing CD, the feature maps from different time instances are subjected to the channel exchange procedure. This design facilitates the integration of temporal and spatial features at shallower layers of the network. By enabling interchannel communication, the channel exchange mechanism expands the representational capacity of the model and supports the extraction of more comprehensive features relevant to CD.

As shown in Fig. 4, the composite representation attention (CRA) module is a core part of the multiscale feature extraction process. It consists of two separate attention branches with independent weights, each processing one of the dual-time-series input tensors. The proposed module incorporates both channel and spatial attention mechanisms to capture complementary information across feature dimensions, while residual connections are employed to maintain the integrity of the original representations.

The CRA process can be formulated as follows:

$$\mathbf{T}_1 = \mathbf{T}_1 \odot \text{Atten}_{T1}^{\text{channel}}(\mathbf{T}_1) \odot \text{Atten}_{T1}^{\text{spatial}}(\mathbf{T}_1) \quad (1)$$

$$\mathbf{T}_2 = \mathbf{T}_2 \odot \text{Atten}_{T2}^{\text{channel}}(\mathbf{T}_2) \odot \text{Atten}_{T2}^{\text{spatial}}(\mathbf{T}_2) \quad (2)$$

$$\textbf{Output} = \mathbf{T}_1 + \mathbf{T}_2. \quad (3)$$

1) *Channelwise Attention Mechanism:* This mechanism captures interchannel dependencies [25] through adaptive max pooling and two successive 2-D convolutional layers with ReLU activation, ultimately producing channel attention weights via a Sigmoid function.
2) *Spatial Attention Mechanism:* To capture spatial dependencies [26], the network applies vector max pooling and a subsequent 1-D convolution, with a Sigmoid function used to generate spatial attention maps that highlight informative regions.

b) Composite Representation Attention: In the CRA module, feature tensors from different time points are processed through their respective attention branches, then combined via elementwise addition. This fusion produces a composite feature representation that incorporates both spatial and channelwise contextual information, which can be used in subsequent processing stages.

c) Multiscale Aggregation: As shown in Fig. 5, the multiscale aggregation (MSA) [27] is composed of four parallel convolutional streams and one skip connection. Each stream includes convolutional layers with varying kernel sizes, designed to process the input feature maps at multiple receptive fields.

Specifically, the multistream module consists of four parallel convolutional streams.

1) *Stream 1:* Consecutive convolutional layers with kernel sizes of $(1, 3)$ and $(3, 1)$.
2) *Stream 2:* Consecutive convolutional layers with kernel sizes of $(1, 5)$ and $(5, 1)$.
3) *Stream 3:* Consecutive convolutional layers with kernel sizes of $(1, 7)$ and $(7, 1)$.
4) *Stream 4:* A single convolution with a kernel size of $(1, 1)$.

These combinations allow the network to encode features with different spatial extents. In parallel, a skip convolution layer with a (1, 1) kernel and batch normalization is used to preserve the original input information.

The outputs of all streams are concatenated and followed by an elementwise matrix multiplication operation. This interaction between processed and original features allows the network to construct a composite multiscale feature map.

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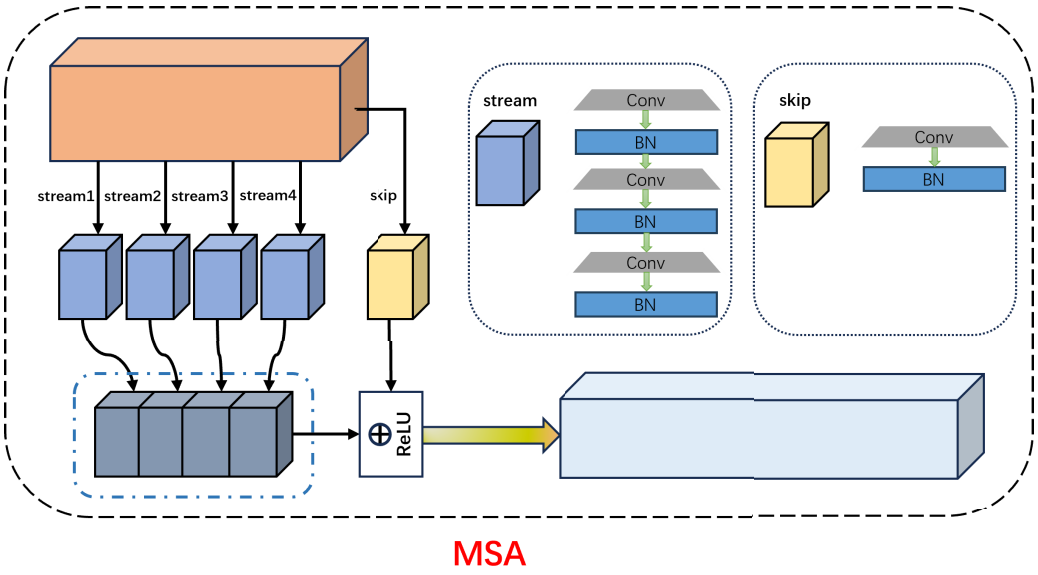


Fig. 5. Illustration of the proposed MSA module.

Subsequently, the combined attention responses are processed through a Sigmoid activation layer, producing outputs that serve as inputs for subsequent network components. Thus, the MSA can be denoted as follows:

$$\mathbf{B}_i = \text{Conv2d}\left(C_{\text{in}}, C_{\text{out}}, 1\right) \quad \forall i \in \{0,1,2,3\} \tag{4}$$

$$\mathbf{B}_i = \mathbf{B}_i \odot \sum_{k\in\{1,3,5,7\}} \text{Conv2d}\left(C_{\text{out}}, C_{\text{out}}, (k,k), \text{padding} = \frac{k}{2}, \text{dilation} = k\right) \tag{5}$$

$$\mathbf{X}_{\text{cat}} = \text{Conv2d}\left(4C_{\text{out}}, C_{\text{out}}, 3, \text{padding} = 1\right)\left(\sum_{i=0}^{3} \mathbf{B}_i\right) \tag{6}$$

$$\mathbf{X} = \text{ReLU}\left(\mathbf{X}_{\text{cat}} + \text{Conv2d}\left(C_{\text{in}}, C_{\text{out}}, 1\right)(x)\right). \tag{7}$$

d) Multiscale Fusion: As shown in Fig. 6, the MF is responsible for combining feature maps from different levels [28]. This module includes multiple upsampling modules, convolution–normalization modules, and a projector. In particular, the input comprises one high-level feature map E and two low-level feature maps C and D. Initially, feature map E is upsampled and passed through a convolution layer to extract high-level semantic information. The high-level feature map E is subsequently combined with C and D via elementwise multiplication to integrate features across multiple scales.

This process reduces the impact of noise on the model's learning while ensuring that semantic information from different scales is learned. Thereafter, an upsampling–convolution–normalization operation is applied to preserve the structural details of E, and the resulting feature map is concatenated with the fusion output of E and D. Finally, the concatenated feature maps are processed through a convolutional layer with a (3×3) kernel, followed by a (1×1) projection layer, producing a single-channel representation vector map (RVM). This vector map contains the fused

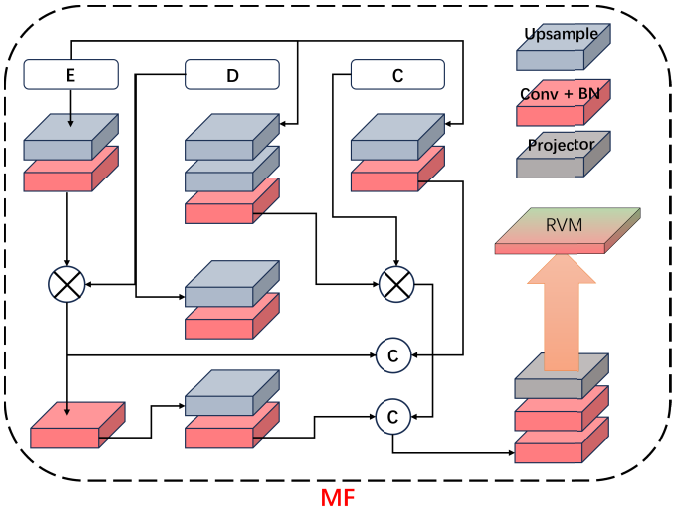


Fig. 6. Illustration of the proposed MF module.

multilevel semantic information from feature maps C, D, and E. Therefore, the MF can be denoted as follows:

$$\mathbf{x}_1' = \mathbf{x}_1 \tag{8}$$

$$\mathbf{x}_2' = \sum_{i=1}^{2} f_i\left(\mathbf{x}_1\right) \cdot \mathbf{x}_2 \tag{9}$$

$$\mathbf{x}_3' = \sum_{i=1}^{3} f_i\left(\mathbf{x}_1\right) \cdot f_i\left(\mathbf{x}_2\right) \cdot \mathbf{x}_3 \tag{10}$$

$$\mathbf{x}_2'' = g_1\left(\mathbf{x}_2', g_2\left(f_1\left(\mathbf{x}_1\right)\right)\right) \tag{11}$$

$$\mathbf{x}_3'' = g_1\left(\mathbf{x}_3', g_2\left(f_1\left(\mathbf{x}_2''\right)\right)\right) \tag{12}$$

$$\textbf{Output} = h_1\left(h_2\left(\mathbf{x}_3''\right)\right). \tag{13}$$

e) Representation-Guided Decoder: As shown in Fig. 7, the representation-guided decoder (RGD) is responsible for

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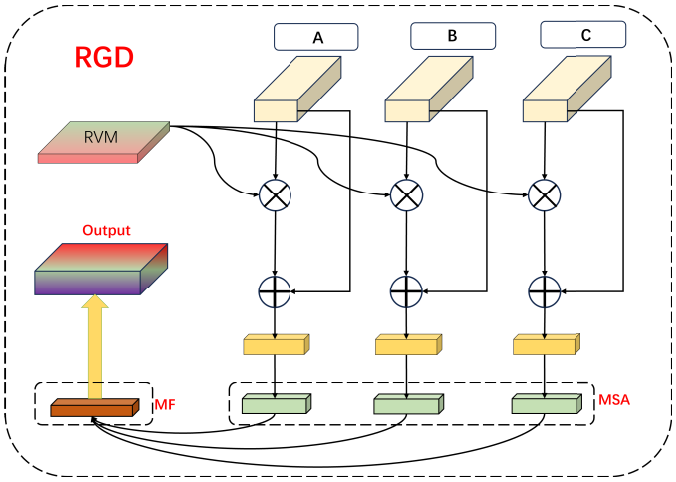


Fig. 7. Illustration of the proposed RGD module.

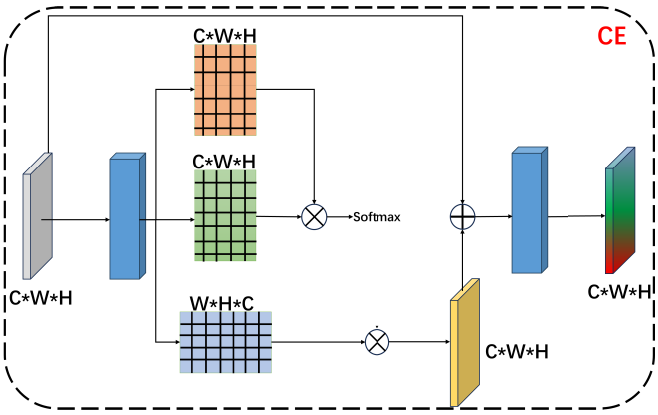


Fig. 8. Overview of the proposed CE module.

merging feature maps from both deep and shallow layers of the network. The input feature maps include the deep-layer feature vector map C, the RVM from the MF layer, and the shallow-layer feature maps A and B. The three feature maps are first multiplied with the RVM output and then combined through elementwise addition with the initial input feature maps A, B, and C, producing output feature maps that retain the original input dimensions.

These feature maps are processed by the MSA module, which enhances the internal spatial and channel dependencies of each image. Finally, the MF layer fuses the features, producing a globally guided semantic feature map. This feature map not only retains both low- and high-level semantic information but also effectively combines features from different scales, ensuring accurate CD.

5) CE: As illustrated in Fig. 8, the module integrates a channel attention mechanism (CAM) that emphasizes informative channels within the feature representation [29]. In the CE module, the input feature map is first reshaped into a flattened form, after which a dot-product operation between the query and key vectors is performed to obtain the correlation matrix. This result is subsequently normalized using a Softmax function to generate the attention distribution. The feature map is then weighted according to the attention values. The gamma parameter controls the fusion of the input feature map with the weighted result. The CE module further refines the input representation through a sequence of convolutional layers designed to strengthen its feature discrimination capability. The process can be formulated as follows:

$$\mathbf{Q} = \text{view}(x), \quad \mathbf{K} = \text{view}(x)^T \tag{14}$$

$$\text{Energy} = \mathbf{Q} \cdot \mathbf{K} \tag{15}$$

$$\text{Attention} = \text{Softmax}(\text{Energy}) \tag{16}$$

$$\mathbf{Out} = \gamma \cdot (\text{Attention} \cdot \mathbf{V}) + x. \tag{17}$$

6) Loss Calculation: To effectively train the network for CD, the loss function is formulated to account for both fine-grained and holistic change cues in remote sensing images. Concretely, the overall loss combines two terms: one measuring the discrepancy between the predicted change map (prediction) and the GT (label), and another assessing the difference between the multiscale fused RVM and the same GT. The total loss is therefore expressed as follows:

$$L_{\text{total}} = L_{\text{prediction}} + L_{\text{RVM}} \tag{18}$$

where

1) The first term BCE(prediction, label) is the binary cross-entropy loss between the predicted CD map prediction and the GT label, ensuring pixel-level accuracy in detecting changes.
2) The second term BCE($RVM, label$) calculates the BCE loss between the multiscale fused RVM and the GT label. The RVM aggregates multiscale features from both fine-grained local and global contexts, enabling the model to detect larger scale changes that might not be captured by pixel-level analysis alone.

All components of the loss are computed using the binary cross-entropy (BCE) criterion, which is commonly applied for binary classification problems, including remote sensing CD. The BCE loss can be expressed as follows:

$$\text{BCE}(x, y) = -\left[y \cdot \log(\sigma(x)) + (1 - y) \cdot \log(1 - \sigma(x))\right] \tag{19}$$

x represents the logits output by the model (before applying the sigmoid activation); $\sigma(x)$ is the sigmoid activation function, which transforms the logits into probabilities in the range $(0, 1)$; y is the binary GT label, indicating whether the pixel has changed ($y = 1$) or not ($y = 0$); and $\log(\cdot)$ denotes the natural logarithm.

The BCE loss function calculates the discrepancy between the predicted probability $\sigma(x)$ and the true label y, penalizing the model for incorrect predictions. By minimizing this loss, the model learns to predict accurate probabilities for both pixel-level changes (through the *prediction*) and larger scale, multiscale changes (through the *RVM*).

This design reflects the multiscale nature of the CD problem, where both fine-grained and coarse-grained information are important for accurate change identification.

The BCE loss function is particularly effective for binary classification tasks, making it an ideal choice for CD, where the goal is to classify each pixel as either changed or unchanged. The use of the BCE loss encourages the model to

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produce accurate probabilities, which are crucial for precise detection of changes in both local and global contexts.

In addition to its effectiveness in pixel-level classification, BCE loss is numerically stable and well-suited for the optimization process, ensuring that the model can learn efficiently and converge to an optimal solution. By combining local and global losses, our approach ensures that the model can effectively capture both detailed changes and broader, high-level changes in the remote sensing images, improving the model's overall robustness and performance in real-world applications.

7) Comparison Models:

a) FC-EF [35]: Daudt et al. presented a U-Net-based fully convolutional early-fusion model in which the network receives both temporal images as a single multichannel input. This configuration allows the model to jointly learn spatial and spectral correlations from the paired images, facilitating accurate CD.

b) FC-Siam-Conc [35]: Daudt et al. also introduced a fully convolutional Siamese concatenation network for CD, which employs skip connections to link encoder features from both branches with their corresponding decoder layers. This structure promotes efficient information flow and enhances the model's sensitivity to fine-grained variations between image pairs.

c) FC-Siam-Diff [35]: Daudt et al. further proposed a Siamese difference model that computes the absolute difference of the feature maps generated by the two encoder paths, which is then merged with the decoder via skip connections, enhancing the network's ability to distinguish changed regions.

d) HANet [23]: A discriminative Siamese network designed to address CD tasks. It integrates multiscale features through a hierarchical attention mechanism (HAN), which refines feature representations by employing a lightweight self-attention approach. This mechanism helps capture both local and global context, improving detection accuracy, particularly in complex scenarios.

e) HCGMNet [33]: Han et al. presented a hierarchical change-guided mapping network tailored for very-high-resolution remote sensing imagery, leveraging multilevel contextual cues to refine the delineation of change areas. The model utilizes hierarchical convolution operations to extract multiscale features, progressively merging them to enhance both global and local information. A change guide module (CGM) further refines edge features by leveraging self-attention for improved detection performance.

f) SGSLN [34]: Zhao et al. proposed a binary CD framework featuring an exchanging dual encoder–decoder structure that integrates semantic guidance with spatial localization to enhance both accuracy and interpretability of change representations. This model addresses the interference between features from two temporal images in the multiple encoder–single decoder (MESD) framework by integrating these features at the decision stage. It also applies semantic features to accurately identify changed regions, significantly improving CD performance.

g) SNUNet [37]: Fang et al. proposed a Siamese network with dense connections aimed at preserving localization details in the deeper layers of convolutional networks. It incorporates a channel attention ensemble that highlights the most relevant features across different semantic levels.

h) MSPSNet [38]: Guo et al. proposed a deep multiscale Siamese network that combines PCS and SA mechanisms to capture features at various spatial scales, enhancing the representation of subtle changes.

i) BIT [39]: Chen et al. presented a bitemporal image transformer network that effectively models spatial–temporal relationships by exploiting the transformer's capability to encode contextual dependencies across different temporal inputs.

j) Change Former [36]: Bandara and Patel proposed a Siamese architecture built on transformer principles, combining a hierarchical encoder with a multilayer perceptron decoder to effectively capture multiscale features and long-range dependencies.

k) RSP-BIT [40]: Wang et al. enhanced the BIT model by applying remote sensing-specific (RSP) pretraining, using the MillionAID dataset to initialize the ViTAE network and boost performance in CD tasks.

l) SEIFNet [47]: Huang et al. introduced SEIFNet, a network integrating spatiotemporal feature enhancement and interlevel fusion, aimed at improving the identification of changes in satellite and aerial images. It employs a Siamese backbone combined with difference enhancement modules and an adaptive fusion decoder to effectively improve feature representation. Experimental evaluations indicate that SEIFNet surpasses eight contemporary state-of-the-art approaches on multiple benchmark datasets, demonstrating notable improvements in both accuracy and computational efficiency.

8) Evaluation Metrics: The model's performance is assessed using a set of widely adopted evaluation metrics, including the following.

1) *F1-score (F1):* The $F1$-score, defined as the harmonic mean of precision and recall, offers a balanced assessment of the model's performance in correctly identifying positive cases. It is calculated as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{20}$$

2) *Precision:* Precision quantifies the fraction of predicted positives that are true positives, reflecting the model's reliability in positive predictions. It is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

where TP denotes true positives and FP denotes false positives.

3) *Recall:* Recall indicates the fraction of actual positive samples correctly identified by the model, assessing its sensitivity. It is computed as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{22}$$

where FN represents false negatives.

4) *Overall Accuracy:* Overall accuracy (OA) measures the proportion of correctly classified pixels, both positive

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TABLE II
QUANTITATIVE RESULTS OF COMPARED BASELINE METHODS ON THE FCDD (INCREMENT OVER BASELINE)

Method	F1 (%)	IoU (%)	Precision (%)	Recall (%)	OA (%)
Baseline	12.58	6.71	22.10	8.79	92.45
Baseline+MSA+MF	48.61 (+36.03)	32.11 (+25.40)	50.98 (+28.88)	46.44 (+37.65)	**93.93 (+1.48)**
Baseline+MSA+MF+CRA	77.27 (+64.69)	62.96 (+56.25)	**78.71 (+56.61)**	75.88 (+67.09)	90.73 (-1.72)
Baseline+MSA+MF+CRA+CE	77.58 (+65.00)	63.38 (+56.67)	77.32 (+55.22)	77.85 (+69.06)	90.66 (-1.79)
Baseline+MSA+MF+CRA+CE+ChannelExchange	**78.51 (+65.93)**	**64.62 (+57.91)**	**78.71 (+56.61)**	**78.30 (+69.51)**	91.10 (-1.35)

and negative, relative to the total number of pixels. It is defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{23}$$

where TN denotes true negatives.

The evaluation metrics are calculated by contrasting the predicted CD maps with the GT, offering a comprehensive measure of the model's effectiveness in detecting land cover alterations, especially for forested areas.

The above description highlights the setup and configuration of our experimental environment, dataset, and evaluation methodology.

III. EXPERIMENTS AND RESULTS

The experimental setup is designed to benchmark the proposed model against current state-of-the-art methods. In addition, ablation studies are conducted to quantify the contribution of individual components within the network.

A. Design of Experiments

The network is implemented in PyTorch and trained on an NVIDIA RTX 4090 GPU. Model parameters are optimized with AdamW, setting the learning rate to 5×10^{-4} and weight decay to 2.5×10^{-3}. A `CosineAnnealingWarmRestarts` scheduler is applied to adapt the learning rate throughout training. Training proceeds for 100 epochs with a batch size of 16 and 12 data-loading workers. Model performance is evaluated after each epoch, and the version with the best validation accuracy is selected as the final model.

Overall, the training setup ensures efficient training by using GPU acceleration, batch processing, and dynamic learning rate adjustment, with a focus on maximizing model performance on the FCDD.

Throughout the experiments, we ensured that all models were trained under the same hardware environment and used the publicly available implementations and default hyperparameters from GitHub to ensure a fair comparison with other methods. By quantitatively comparing FCDNet with existing methods, we have demonstrated its superiority in remote sensing image CD, particularly in feature extraction and detection accuracy. FCDNet exhibits significant improvements in performance indicators such as $F1$ and OA, while also showing robust generalization across diverse datasets.

B. Ablation Study

The impact of individual modules in the proposed architecture was examined through experiments using a baseline VGG16–U-Net network. This baseline represents the foundational network structure without incorporating any of the designed modules, ensuring a robust and reliable starting point for assessing the contribution of each enhancement.

Table II shows the results of various module arrangements on the FCDD dataset, highlighting the incremental improvements contributed by each component.

Introducing the MSA and MF modules substantially improved model performance. The $F1$-score increased to 48.61% (+ 36.03), IoU rose to 32.11% (+ 25.40), Precision improved to 50.98% (+ 28.88), Recall reached 46.44% (+ 37.65), and OA increased slightly to 93.93% (+ 1.48). The results indicate that these modules improve the extraction of diverse, multiscale features, which play a critical role in accurate forest CD.

Adding the CRA module further refined the model's performance. The $F1$-score increased significantly to 77.27% (+ 28.66), IoU improved to 62.96% (+ 30.85), Precision rose to 78.71% (+ 27.73), Recall reached 75.88% (+ 29.44), and OA achieved 90.73% (+ 3.28). In particular, the CRA module significantly improves the model's ability to align and integrate features across varying resolutions.

The inclusion of the CE module provided incremental improvements, particularly in Recall. The $F1$-score reached 77.58% (+ 65.00), IoU rose to 63.38% (+ 56.67), Recall improved to 77.85% (+ 69.06), while OA slightly decreased to 90.66% (−1.79), and Precision remained stable at 77.32% (+ 55.22). These results confirm that the CE module effectively enhances feature representation, particularly in regions of change.

Finally, incorporating the channel exchange module yielded the best performance across all metrics. The $F1$-score peaked at 78.51% (+ 65.96), IoU increased to 64.62% (+ 57.91), Precision improved to 78.71% (+ 56.61), Recall rose to 78.30% (+ 69.51), and OA achieved 91.10% (−1.35). These results highlight the synergistic effect of combining the CE and channel exchange modules in optimizing feature interaction and improving both recall and precision.

Since the baseline and Baseline+MSA+MF's metrics are poor, for the validity of the analysis, we only compare the other three methods' visual results with Figs. 9–11.

Fig. 9 illustrates the results of the ablation study conducted on the FCDD dataset, showcasing the comparative performance of the three examined models: Baseline + MSA + MF + CRA, Baseline + MSA + MF + CRA + CE, and Baseline + MSA + MF + CRA + CE + ChannelExchange. For illustration, we selected several challenging samples (a)–(f) from the test set, which include complex building features and difficult detection scenarios. These visual examples offer

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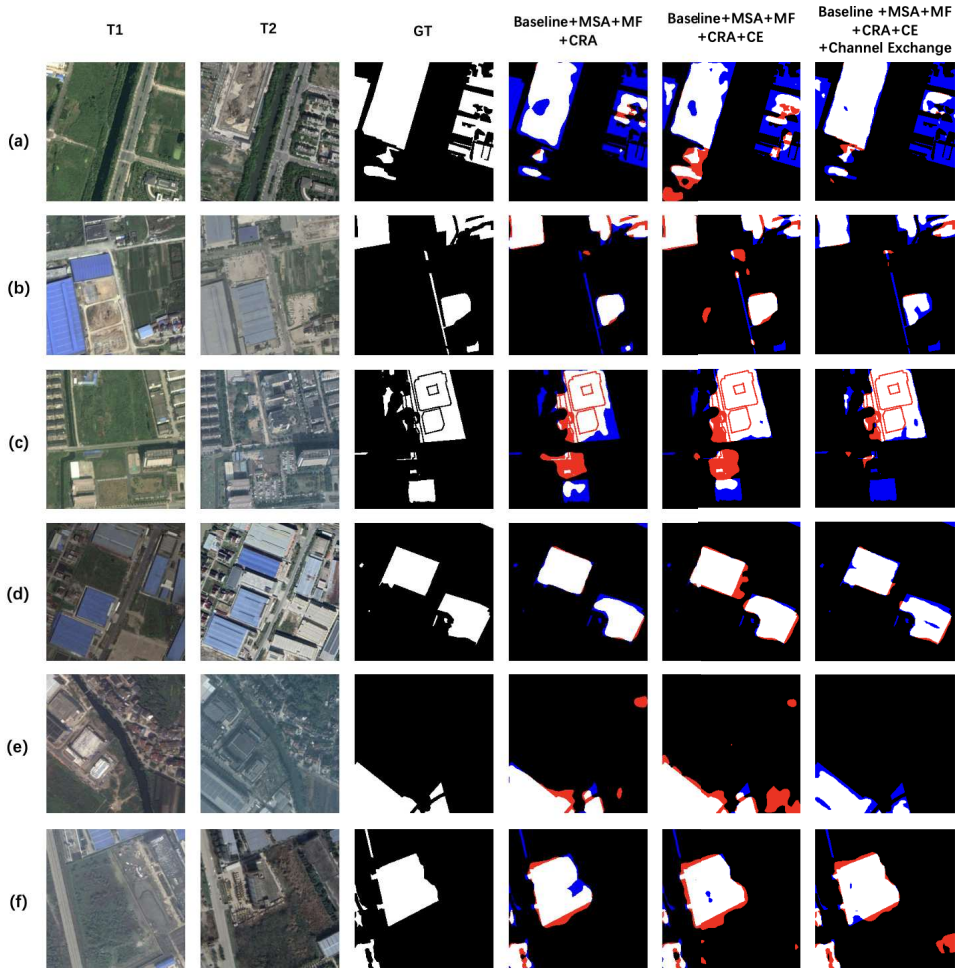


Fig. 9. Visual comparisons of various methods on the forest CD test sets are presented, with distinct colors employed to enhance interpretability: FP in red, TP in white, TN in black, and FN in blue.

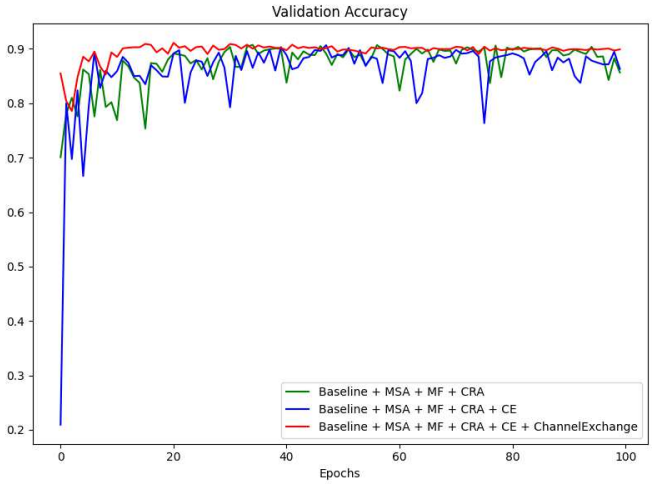


Fig. 10. Validation Acc of each epoch.

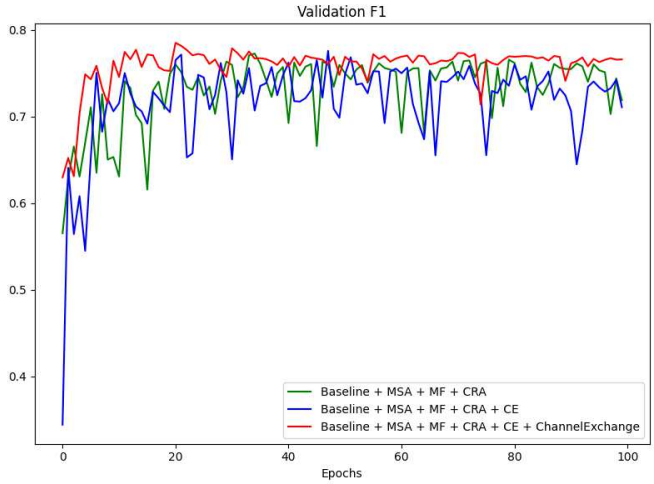


Fig. 11. Validation $F1$ of each epoch.

a clear depiction of the differences in performance among the models. To enhance the intuitiveness of this visualization, different colors are employed, with small red boxes highlighting key detection areas. Blue areas represent missed detections, while red areas indicate false positives.

First, the model Baseline + MSA + MF + CRA exhibits noticeable shortcomings in handling false negatives, especially in complex building features. For example, in sample (a), (b), and (e), we observe significant missed detections where the

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model fails to correctly identify the boundaries of buildings. The model's limited capability to refine small-scale features leads to these errors. As we add CE, the model's performance improves. Specifically, in sample (d), we notice fewer missed detections (fewer blue pixels), particularly in regions where small semantic features of buildings are present.

Next, the inclusion of ChannelExchange in the model, as seen in the third configuration, Baseline + MSA + MF + CRA + CE + ChannelExchange, brings the most significant improvements. This model effectively addresses both false negatives and false positives. In samples (c)–(f), where complex background changes such as seasonal variations and land use cover changes are evident, the model performs much better at distinguishing between irrelevant changes and actual building modifications. In contrast, Baseline + MSA + MF + CRA and Baseline + MSA + MF + CRA + CE models show higher sensitivity to nonbuilding changes, causing false detections (highlighted in red) in regions like trees and roads.

Additionally, Baseline + MSA + MF + CRA + CE + ChannelExchange shows superior performance in detecting fine-grained building details, as seen in sample (f). Even in irregular building shapes, where the building edges are not perfectly aligned with conventional rectangular shapes, this model is much more capable of capturing detailed features compared to the other configurations. This is particularly evident in sample (f), where the irregular building's edges are more accurately delineated with fewer false positives.

As shown in Figs. 10 and 11, the $F1$ and Acc start from the Baseline + MSA + MF + CRA model and gradually improve as the CE module is introduced, ultimately reaching their highest values when the channel exchange module is added. This trend further demonstrates the significant role of these module improvements in enhancing model performance, especially in CD tasks, where the $F1$ and Acc increase indicate higher accuracy and stability.

In summary, comparisons with the baseline network highlight the specific contributions of each module to the overall model performance. The CE module effectively improves recall of change regions, while the channel exchange module plays a key role in enhancing precision and overall performance. These improvements have enabled our model to achieve superior CD results on the FCDD, further demonstrating the potential of these modules in enhancing model robustness and accuracy. Future work may investigate the integration and refinement of these modules to further enhance the network's adaptability and CD accuracy in more challenging environments.

C. Comparison With State-of-the-Art Methods

The analysis in Table III shows that FCDNet achieves superior results compared with several leading methods—such as FC-EF, FC-Siam-conc, HANet, and SGSLN—across key evaluation indicators, including $F1$, IoU, Precision, Recall, and Accuracy. Table III demonstrates that FCDNet surpasses current approaches, attaining an impressive $F1$-score of 78.51% on the FCDD dataset.

TABLE III
QUANTITATIVE RESULTS OF COMPARED METHODS ON THE FCDD

Method	F1 (%)	Pre. (%)	Rec. (%)	Acc. (%)	IoU (%)
FC-EF	71.17	69.68	72.72	87.77	55.24
FC-Siam-conc	72.04	72.97	71.14	88.54	56.30
FC-Siam-diff	60.08	62.29	58.03	83.99	42.94
HANet	55.90	57.32	54.54	82.14	38.79
HCGMNet-CD	74.89	78.13	71.92	89.99	59.86
SEIFNet	75.84	77.43	74.32	90.17	61.09
SGSLN	75.96	75.79	76.13	90.00	61.23
FCDNet(Ours)	78.51	78.71	78.30	91.10	64.62

TABLE IV
QUANTITATIVE RESULTS OF EIGHT COMPARED METHODS AND OURSS ON THE LEVIR-CD DATASET

Method	F1 (%)	Pre. (%)	Rec. (%)	OA (%)	IoU (%)
FC-EF	61.52	73.31	53.00	-	44.43
FC-Siam-diff	89.00	91.76	86.40	-	80.18
SNUNet	89.97	91.31	88.67	98.99	81.77
MSPSNet	89.67	90.75	88.61	98.96	81.27
BIT	89.94	90.33	89.56	98.98	81.72
Change Former	90.20	92.05	88.37	99.01	82.21
RSP-BITs	89.71	92.00	87.53	98.98	81.34
HANet	90.28	91.21	89.36	99.02	82.27
FCDNet(Ours)	90.48	92.10	88.92	99.05	82.61

This enhanced performance is primarily due to FCDNet's capacity to effectively aggregate contextual semantic information across different spatial scales. The HAN module integrated into FCDNet strengthens the network's capacity to detect fine-grained and precise changes, effectively handling complex forest environments with seasonal and interannual variations.

Fig. 12 presents visual comparisons on the FCDD, highlighting FCDNet's ability to detect both subtle and complex changes with greater accuracy. The qualitative results validate the quantitative improvements and highlight FCDNet's capability to extract detailed semantic information at different spatial scales.

First, FCDNet demonstrates a stronger ability to reduce false negatives when detecting complex forest structures. Compared with attention-based models such as SEIFNet, HANet, and HCGMNet, the visualizations show that FCDNet produces noticeably fewer blue pixels.

Second, FCDNet demonstrates a strong capability to suppress false alarms in challenging areas characterized by intricate background patterns. It effectively distinguishes genuine forest disturbances from pseudo-changes arising from seasonal dynamics or variations in land cover, thereby ensuring higher detection precision in diverse temporal contexts. For instance, in Fig. 12(b) and (e), FCDNet effectively avoids detecting nonbuilding structures, like trees, as changes, a challenge for models such as FC-EF, SGSLN, SEIFNet, HANet, and HCGMNet, which are more sensitive to such irrelevant features.

Third, Tables IV and V summarize the quantitative performance comparisons on the WHU-CD and LEVIR-CD datasets. FCDNet consistently surpasses competing methods on key evaluation metrics, including $F1$-score, OA, and Recall.

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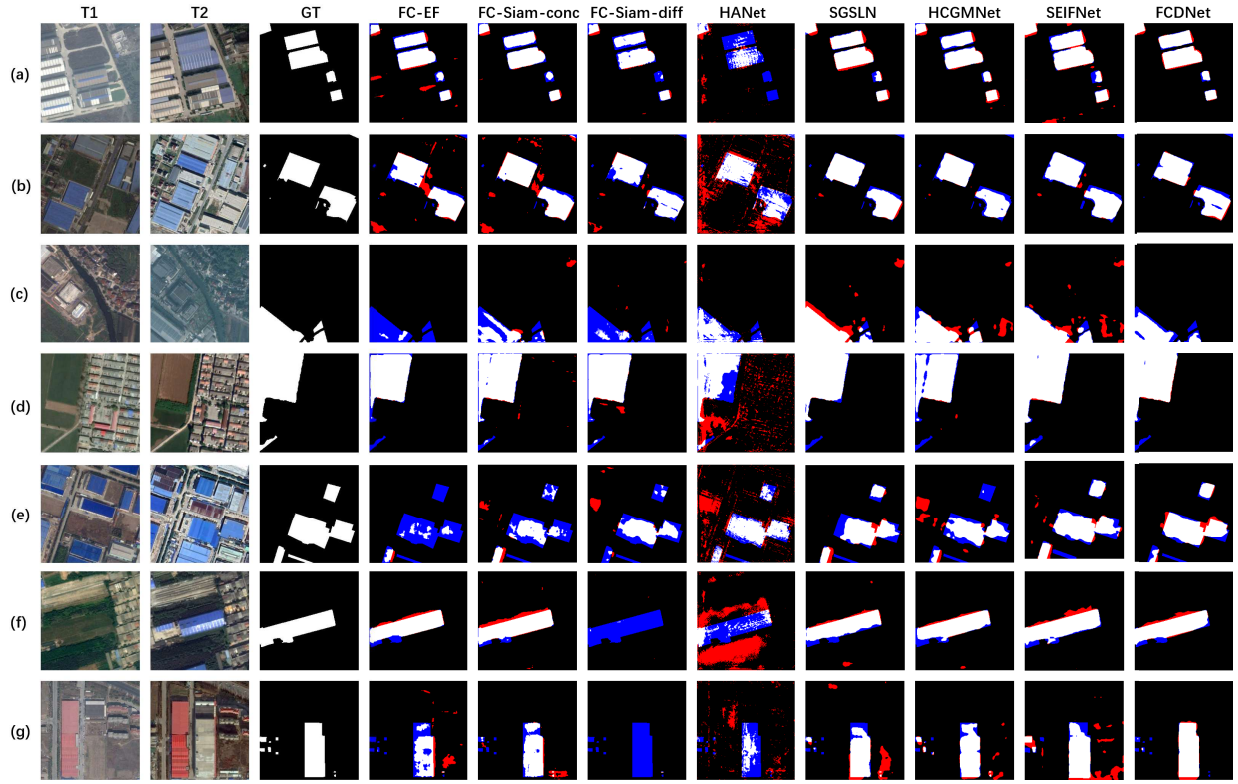


Fig. 12. Overview of the results.

TABLE V
QUANTITATIVE RESULTS OF EIGHT COMPARED METHODS AND OURSS ON THE WHU-CD DATASET

Method	F1 (%)	Pre. (%)	Rec. (%)	OA (%)	IoU (%)
FC-EF	58.05	76.49	46.77	-	40.89
FC-Siam-diff	86.31	89.63	83.22	-	75.91
SNUNet	87.76	87.84	87.68	99.13	78.19
MSPSNet	86.49	87.84	85.17	99.05	76.19
BIT	80.97	74.01	89.37	98.51	68.02
Change Former	87.18	92.70	82.28	99.14	77.27
RSP-BITs	78.50	69.93	89.45	98.26	64.60
HANet	88.16	88.30	88.01	99.16	78.82
FCDNet(Ours)	92.57	94.45	90.77	99.42	86.17

Remarkably, FCDNet attains $F1$-scores of 90.48% on LEVIR-CD and 92.57% on WHU-CD, exceeding HANet by 0.20 and 4.41 percentage points, respectively. This improvement is particularly significant considering that Change Former, a transformer-based model with a larger parameter count, achieves lower scores.

Overall, experiments conducted on the FCDD, LEVIR-CD, and WHU-CD datasets clearly verify the outstanding performance of FCDNet in accurately identifying changes within high-resolution remote sensing data. Whether dealing with complex seasonal changes or addressing highly imbalanced changed and unchanged pixel distributions, FCDNet consistently shows significant advantages. Subsequent studies may investigate extending FCDNet to additional remote sensing datasets, with an emphasis on enhancing computational efficiency and suitability for real-time CD applications.

D. Forest CD Capability Validation

Fig. 13 illustrates the application of our FCDD-trained model to CD samples from a Swedish region, using Sentinel-2 images acquired in 2018 (T1) and 2023 (T2). To assess the model's performance, representative samples were manually annotated, covering a range of complex vegetation dynamics, land use changes, and environmental disturbances. The results highlight the model's exceptional accuracy in identifying changed regions, with particularly low false positive and false negative rates. It demonstrates strong capability in detecting fine-grained changes in challenging conditions, such as sparse vegetation and small-scale land alterations, and shows robust adaptability to various environmental contexts, including seasonal and spectral variations.

IV. DISCUSSION

FCDNet has demonstrated significant advantages in forest CD, especially in dealing with spectral heterogeneity and temporal dynamics, providing more accurate detection results than traditional methods (e.g., spectral disparity and image alignment). Traditional approaches often struggle to handle the complexity of forest ecosystems, particularly under seasonal variations and environmental disturbances, whereas FCDNet significantly improves detection accuracy through multiscale feature fusion and CE mechanisms.

To further analyze the role and learned representations of each component, gradient-weighted class activation mapping (Grad-CAM) was utilized to visualize the intermediate feature maps, as illustrated in Fig. 14. Grad-CAM utilizes

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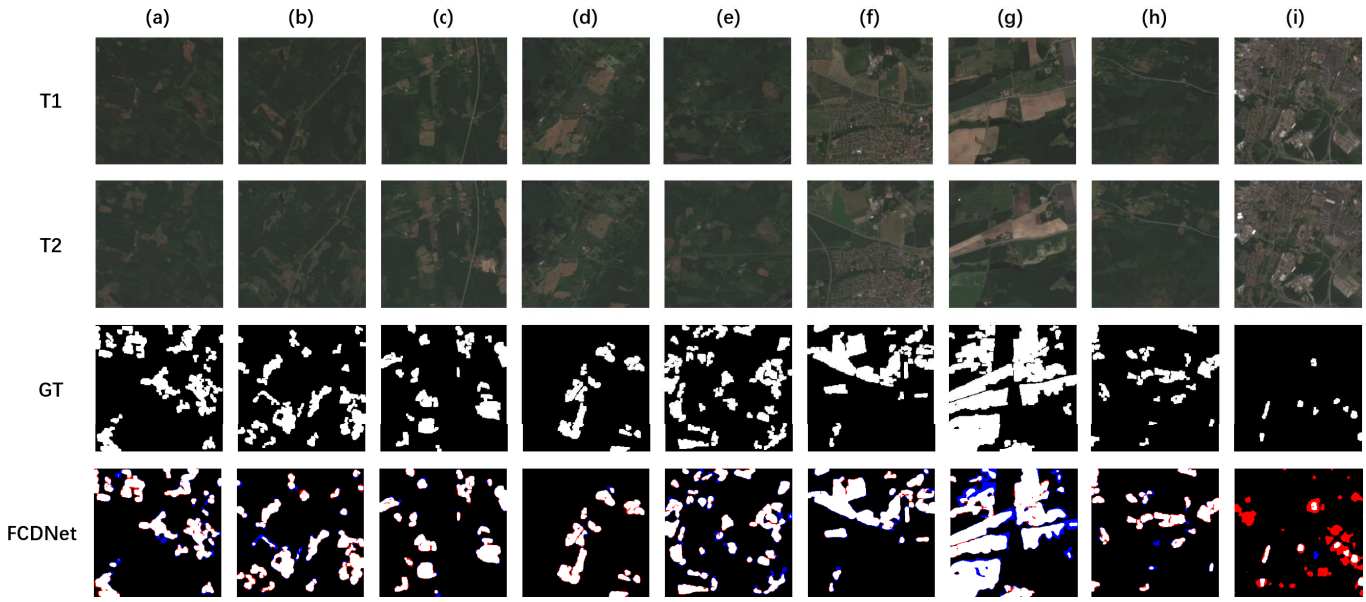


Fig. 13. Visualization of results on Sweden CD samples.

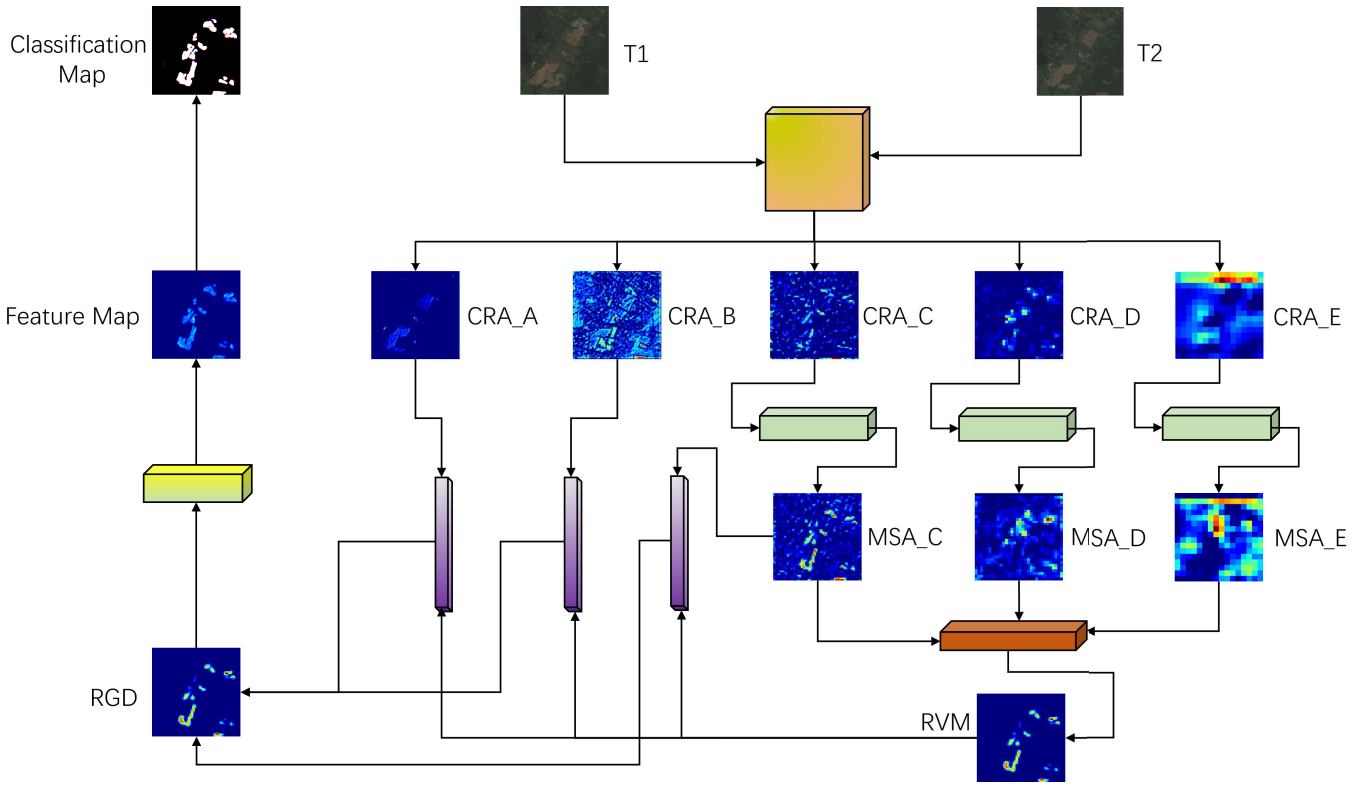


Fig. 14. Visualization process of feature extraction in various stages of FCDNet.

the backpropagated gradients from the final convolutional layers to emphasize the most influential regions of the input, thus improving the interpretability of the model's decision process. As shown in the results, the CRA module performs initial downsampling and feature fusion on images T1 and T2, with CRA_A to CRA_E indicating that forest and nonforest regions are preliminarily distinguished. The MSA module further enhances the spatial boundaries of potential change areas through multiscale context modeling, though some uncertainties remain.

Subsequently, the MF module integrates the MSA features and provides more precise localization of change regions. The RVM module further sharpens the detected boundaries, improving local accuracy. The RGD module introduces global guidance to fine-tune the feature maps, promoting consistent and stable model responses. The final classification map shows

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that the model delivers high accuracy and robustness in identifying forest change areas. These visualizations confirm the synergistic contribution of each component and demonstrate the effectiveness of Grad-CAM in interpreting the model's internal decision-making process.

Nevertheless, FCDNet still faces several challenges. First, its real-time detection capability is limited, particularly in large-scale monitoring scenarios, due to inference speed and computational overhead. Future research may explore model acceleration and hardware optimization for real-time deployment. Second, although the use of Grad-CAM improves interpretability, the overall black-box nature of the decision process still limits user trust in practical applications.

In addition, computational efficiency poses a challenge for large-scale, high-resolution datasets, as the model introduces substantial processing overhead. Approaches like model pruning and knowledge distillation may be employed to reduce the computational burden and enhance efficiency. Finally, generalization remains an important issue. Although FCDNet performs well on standard benchmarks, its robustness in diverse environmental and geographic settings requires further validation.

V. Conclusion

In this work, we present FCDNet, a multiscale attention network specifically designed to detect forest changes in remote sensing images. Through the integration of hierarchical attention, multiscale feature aggregation, and channel interaction strategies, FCDNet achieves notable gains in both detection precision and model robustness. This integrated design allows the network to effectively capture complex forest dynamics, providing a reliable and precise tool for monitoring forest changes. Future work aims to refine the model's capabilities, overcome existing limitations, and explore its use in additional remote sensing CD tasks.

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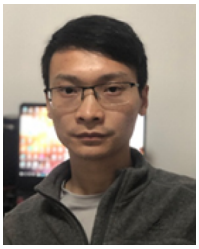
Jun Wang is currently pursuing the bachelor's degree in urban forestry with Beijing Forestry University, Beijing, China.

His research interests include deep learning, change detection, optimization of clustering algorithms, and computer vision.



Zongqi Yao received the B.S. degree in surveying and mapping engineering from China University of Mining and Technology-Beijing, Beijing, China, in 2020. He is currently pursuing the Ph.D. degree in forest management with Beijing Forestry University, Beijing.

His research interests include remote sensing monitoring of forest resources and dynamic changes.



Long Chen received the B.S. degree in geographic information system from Central South University of Forestry and Technology, Changsha, China, in 2016, and the Ph.D. degree in forest management science from Beijing Forestry University, Beijing, China, in 2024.

He is currently an Assistant Professor at the Institute of Forest Resource Information Techniques, Chinese Academy of Forestry, Beijing. His research interests include computer vision, graphic image processing, deep learning, and web development.



Ruijing Yang is currently pursuing the bachelor's degree in information management and information system with Beijing Forestry University, Beijing, China.

Her research interests include deep learning and computer vision.



Xiaoli Zhang (Member, IEEE) received the B.S. degree in computer science from Hebei University, Baoding, China, in 1989, and the Ph.D. degree in forest management science from Beijing Forestry University, Beijing, China, in 1998.

From 1998 to 1999, she was a Post-Doctoral Research Fellow with the Institute of Remote Sensing Application in CAC, Beijing. From 2006 to 2007, she was a Visiting Scholar in remote sensing and intelligent analysis at Lincoln University, Lincoln, New Zealand. She is currently a Full Professor at Beijing Forestry University, Beijing. Her research interests include forest resources and ecological environment remote sensing monitoring and spatial prediction, forest disease and pest remote sensing monitoring and risk assessment, and vegetation quantitative remote sensing.

Prof. Zhang is a Councilor of the Associate on Environment Remote Sensing of China (AERSC), a Supervisor of Beijing Society for Information Technology in Agriculture, and a member of Forest Management Branch of Chinese Society of Forestry. Her awards and honors include The National Scientific and Technological Progress Award, the MOE Scientific and Technological Progress Award, Beijing Science and Technology Award, the Forest Science and Technology Award for Chinese Youth and MOE New Century Excellent Talents (NCET).

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